enabled scientists to turn theoretical concepts into operational technology. The last decade has seen the development of prototype quantum processors from IBM, Google, and Rigetti, which have become accessible through cloud platforms.

QML research began in the mid-2010s when scientists investigated how quantum systems could improve machine learning algorithms. By combining the pattern recognition strengths of ML with the parallelism of quantum computing, they felt they could process extensively complex datasets at extraordinary speeds. Quantum computing parallelism enables systems to evaluate numerous possibilities simultaneously (rather than one by one), which results in faster information retrieval. For instance, the analysis of molecular interactions for new drugs through classical models requires weeks of processing time. Research at the University of Toronto shows that quantum inspired methods immensely accelerate protein structure prediction, a key step in drug design.

Quantum machine learning research continues to expand across the world. In India, research institutions such as IIT Madras and the Indian Institute of Science lead the country's quantum technology efforts. The CQuICC (Centre for Quantum Information, Communication and Computing) at IIT Madras is involved in the research and development of quantum communication systems. The IISc Quantum Technology Initiative (IQTI) was started in 2020. The program promotes quantum research and unites physicists with computer scientists, engineers and material scientists to work on crucial research areas such as quantum computation and simulations, quantum communication and cryptography, quantum sensing and metrology, and quantum materials and devices.

Indian startups are developing practical quantum machine learning systems that can solve real world problems. QpiAI in Bengaluru has secured US$ 32 million (approximately ₹ 279 crore) from Avataar Ventures and India's National Quantum Mission in its Series A funding round. The funding will help QpiAI expand its full stack quantum computing system, which serves the healthcare and manufacturing industries. Bengaluru-based BosonQ Psi has joined IBM's Quantum Network to create quantum algorithms for engineering simulations. The platform enables users to build fast and accurate models of intricate problems faced in the aerospace, automotive and biotech fields. Scientists can perform experiments and proof-of-concept projects using cloud-based quantum tools, which remove the need for costly hardware infrastructure. These startups showcase India's evolution from basic quantum research to practical technology development.

Open source ecosystem for QML

The rapid advancement of quantum machine learning (QML) stems primarily from the open source movement. Open source technology enables cost reduction and fosters worldwide participation in its development. The open nature of quantum computing enables developers, researchers and students to test innovative concepts, share their findings, and build on existing work. Today, innovation in quantum computing is not limited to a few top research labs. It is happening every day on GitHub, and in hackathons and educational institutions.

Multiple worldwide platforms serve as the foundation for this transformation. The quantum computing community relies on three main platforms, which include IBM's Qiskit, Google's Cirq and PennyLane from Canada-based Xanadu. Qiskit is a leading framework for quantum circuit development and simulation. Cirq provides Python-based quantum algorithm development capabilities. PennyLane, which is quite popular with developers these days, connects quantum computing with machine learning. Besides, it also works well with PyTorch and TensorFlow. These platforms demonstrate how open tools make quantum computing available to all users.

India is investing heavily in quantum computing development. The academic programs and research initiatives at IIT Madras and IISc Bangalore include quantum computing as part of their curriculum. The SWAYAM platform hosts IIT Madras's quantum algorithms course, which is also supported by IBM. The students at IISc participate in workshops that teach them to work with tools like PennyLane. Apart from the classroom, communities such as Qiskit India and Quantum Computing India (QCI) are also educating enthusiasts to get involved through hackathons, fellowships and projects. QCI is providing training programs, governance structures and employment opportunities.

The Indian government started the National Quantum Mission (NQM) in April 2023 to encourage the use of quantum computing. The mission's goal is to build quantum computers with 50 to 1000 qubits by 2031, at a budget of ₹60,030 million.

A major breakthrough came in April 2025, when Bengaluru-based startup QpiAI launched QpiAI Indus, a 25-qubit superconducting quantum computer. QpiAI is one of the eight startups supported under the National Quantum Mission. Indus is India's first full stack quantum system. It combines advanced hardware with scalable controls and AI driven quantum software. The system has applications across life sciences, materials science, logistics, climate action and several other fields. Bengaluru serves as home